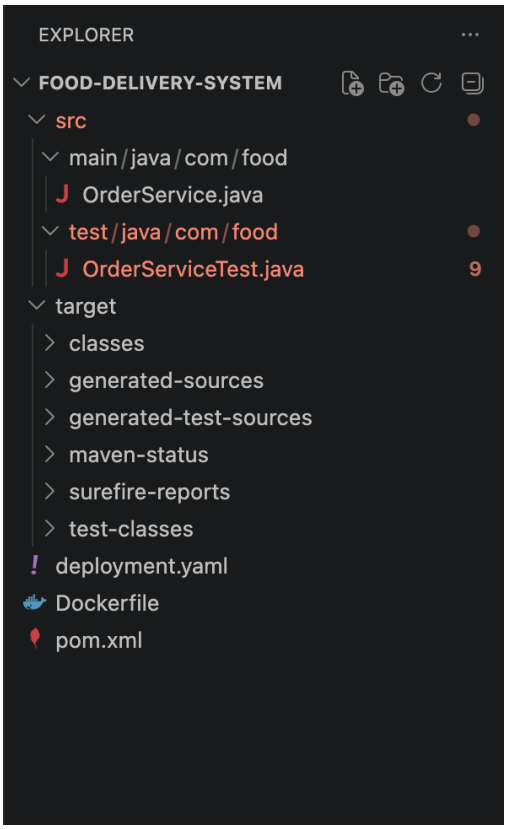
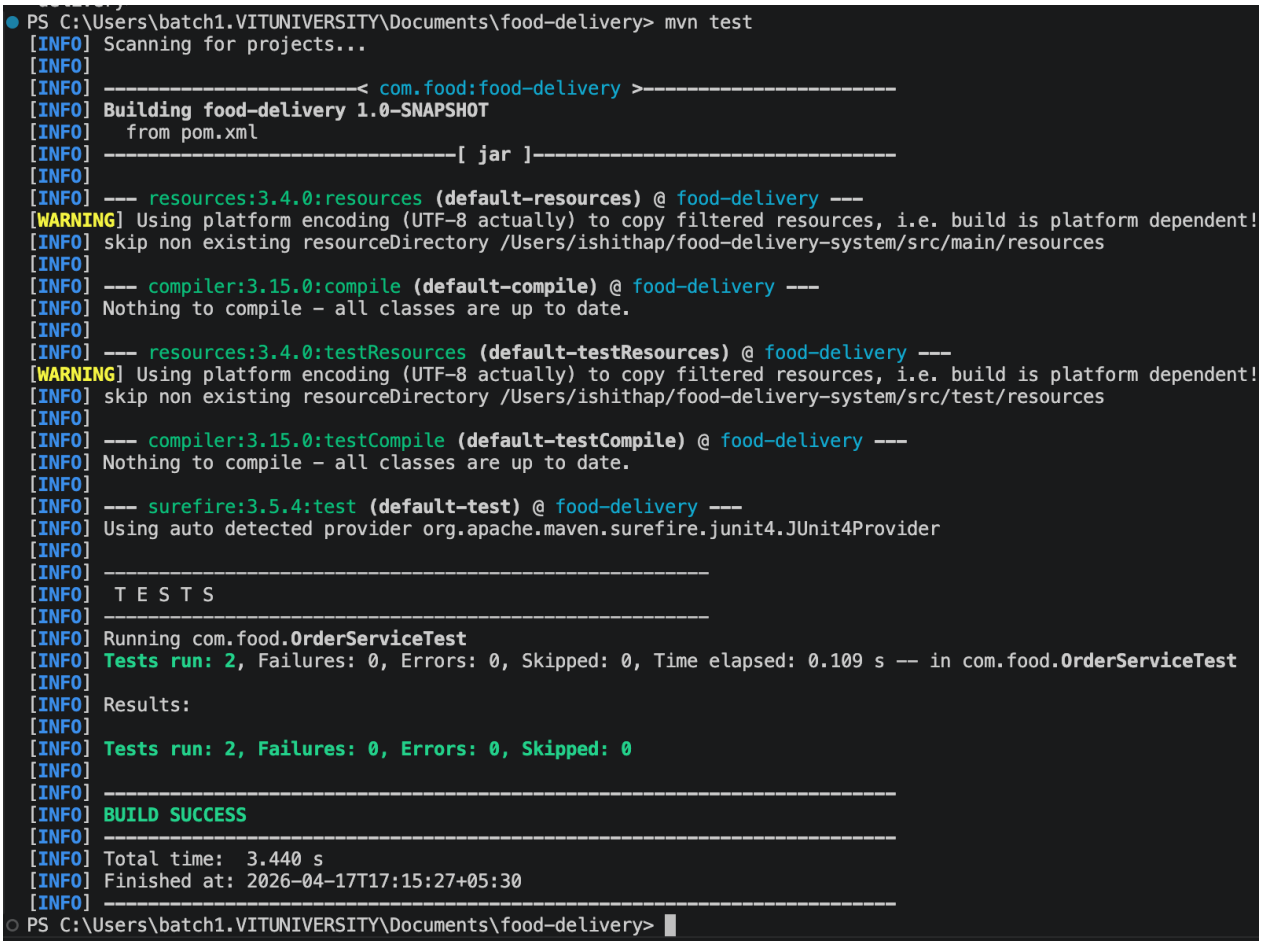
Java Project Structure and Order Management Service Logic in VS Code"

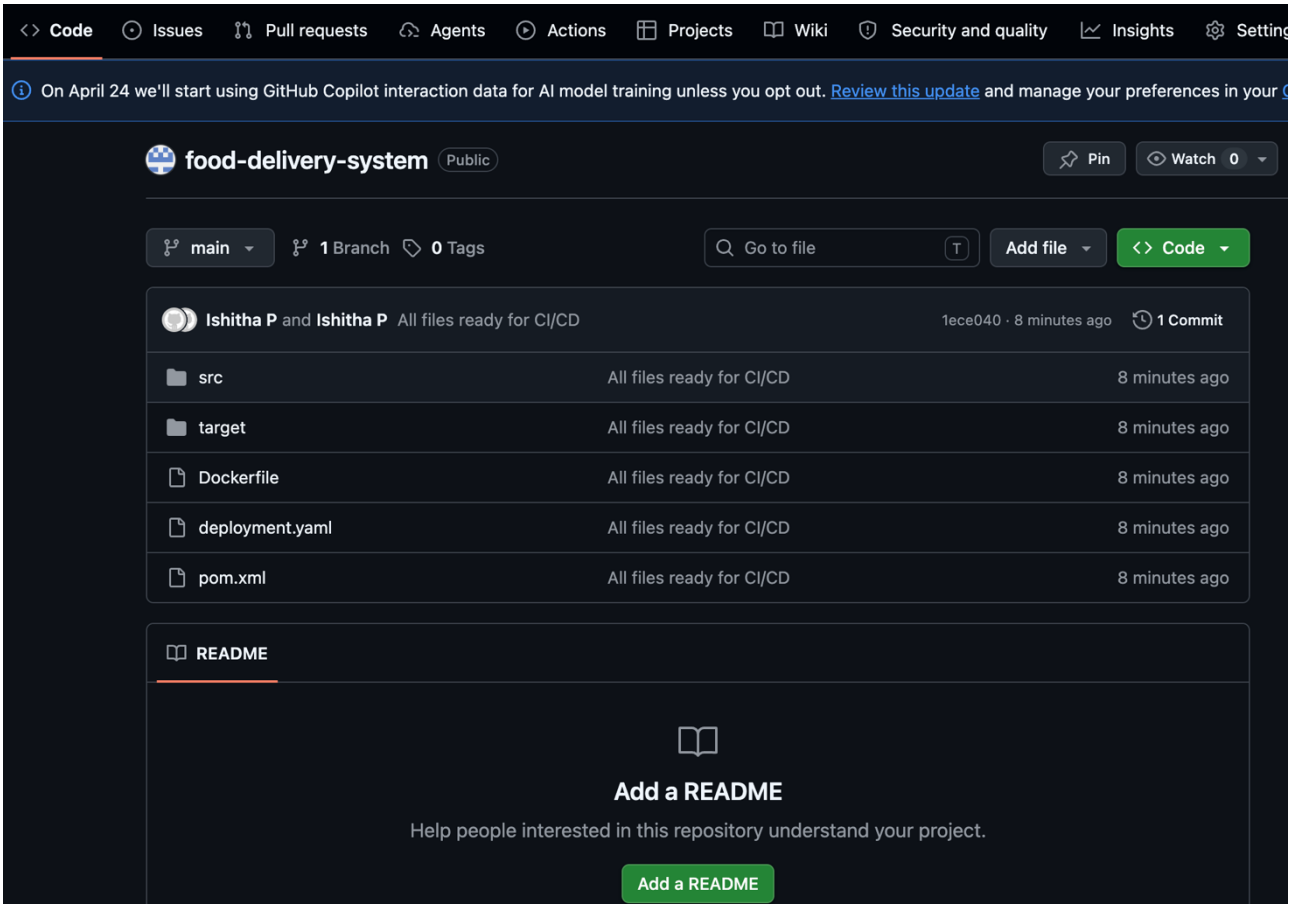


Successful Execution of JUnit Test Cases via Maven Lifecycle

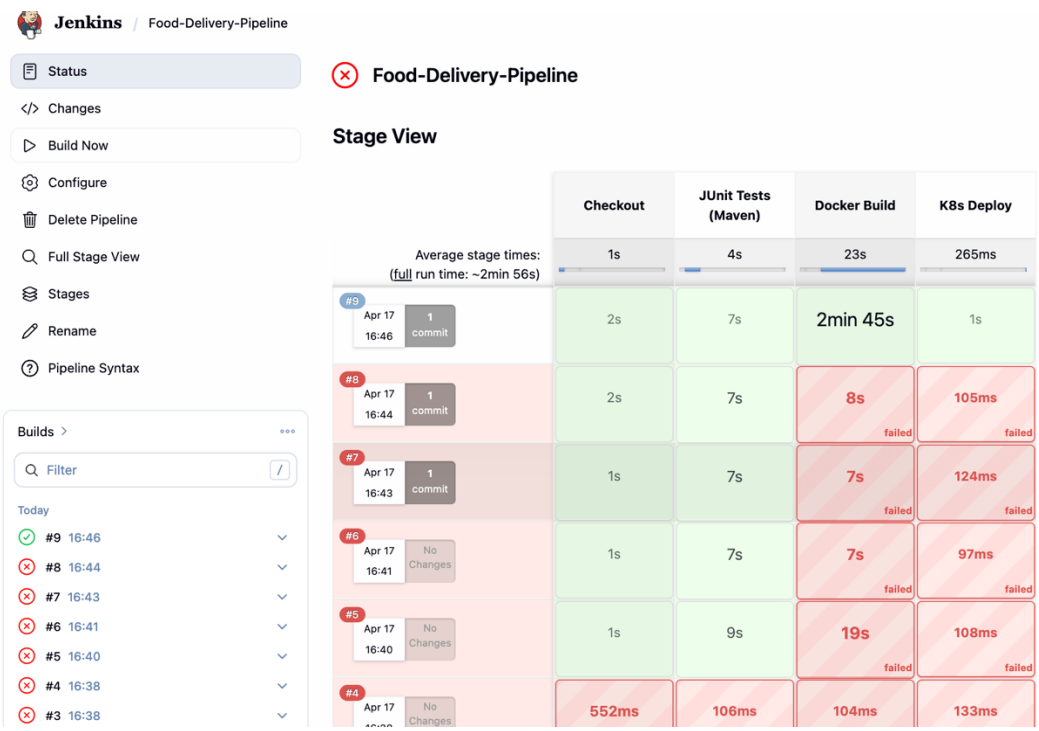


```
PS C:\Users\batch1.VITUNIVERSITY\Documents\food-delivery> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.food:food-delivery >-----------------------
[INFO] Building food-delivery 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ food-delivery ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/ishithap/food-delivery-system/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ food-delivery ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ food-delivery ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/ishithap/food-delivery-system/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ food-delivery ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ food-delivery ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.food.OrderServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.109 s -- in com.food.OrderServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.440 s
[INFO] Finished at: 2026-04-17T17:15:27+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\batch1.VITUNIVERSITY\Documents\food-delivery>
```

Project Repository Hosted on GitHub for Version Control



Jenkins CI/CD Pipeline Stage View with Successful Automation



Verifying Application Deployment in Kubernetes Cluster

```
PS C:\Users\batch1.VITUNIVERSITY\Documents\food-delivery> kubectl get pods
NAME                              READY   STATUS         RESTARTS   AGE
food-delivery-5878f55f69-wj5xg    0/1     ErrImagePull   0          27m
PS C:\Users\batch1.VITUNIVERSITY\Documents\food-delivery>
```

Online Food Delivery System Container Image in Docker Desktop

food-delivery	latest	cdfa711f1e20	2 minutes ago	410.65 MB

Codes:

src/main/java/com/food/OrderService.java

```
package com.food;

public class OrderService {
    public String placeOrder(String item, double price) {
        if (price <= 0) {
            return "Order Failed";
        }
        return "Order Placed: " + item;
    }
}
```

src/test/java/com/food/OrderServiceTest.java

```
package com.food;

import static org.junit.Assert.*;
import org.junit.Test;

public class OrderServiceTest {
    @Test
    public void testOrderSuccess() {
        OrderService service = new OrderService();
        assertEquals("Order Placed: Pizza", service.placeOrder("Pizza", 10.0));
    }

    @Test
    public void testOrderFailure() {
        OrderService service = new OrderService();
        assertEquals("Order Failed", service.placeOrder("Burger", -5.0));
    }
}
```

pom.xml

```
<project xmlns="http://apache.org">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.food</groupId>
    <artifactId>food-delivery</artifactId>
    <version>1.0-SNAPSHOT</version>
```

```
    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>4.13.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>
</project>
```

Dockerfile

```
FROM eclipse-temurin:17-jdk-jammy
COPY target/*.jar app.jar
ENTRYPOINT ["java","-jar","/app.jar"]
```

deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: food-delivery
spec:
  replicas: 1
  selector:
    matchLabels:
      app: food
  template:
    metadata:
      labels:
        app: food
    spec:
      containers:
      - name: food-app
        image: food-delivery:latest
        ports:
        - containerPort: 8080
```

Pipeline script

```
pipeline {
    agent any
    environment {
        PATH = "/usr/local/bin:/usr/bin:/bin:/usr/sbin:/sbin:/opt/homebrew/bin:$PATH"
        KUBECONFIG = "${HOME}/.kube/config"
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url: 'https://github.com'
            }
        }
        stage('JUnit Tests (Maven)') {
```

```
        steps {
            sh 'mvn clean test'
        }
    }
    stage('Docker Build') {
        steps {
            sh 'mvn package -DskipTests'
            sh 'docker build -t food-delivery:latest .'
        }
    }
    stage('K8s Deploy') {
        steps {
            sh 'kubectl --kubeconfig=${KUBECONFIG} apply -f deployment.yaml --validate=false'
        }
    }
}
}
```